Exp.no : 10 **K-Means ALGORITHM**

AIM:

Experiment to understand K-means clustering algorithm for a given data set.

PYTHON CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans

# Read the dataset
df = pd.read_csv("Mall_Customers.csv")

# Display dataset information
print(df.info())

# Display first 5 rows
print(df.head())

# Pair plot
sns.pairplot(df)
plt.show()

# Select Annual Income and Spending Score
features = df.iloc[:, [3, 4]].values

# Apply K-Means clustering with 5 clusters
model = KMeans(n_clusters=5, n_init=10, random_state=42)
model.fit(features)

# Create final dataframe
Final = df.iloc[:, [3, 4]].copy()

# Predict cluster labels
Final["label"] = model.predict(features)

# Display first 5 rows
print(Final.head())
```

```
# Plot the clusters
sns.set_style("whitegrid")

sns.FacetGrid(
    Final,
    hue="label",
    height=8
).map(
    plt.scatter,
    "Annual Income (k$)",
    "Spending Score (1-100)"
).add_legend()

plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.title("K-Means Clustering")
plt.show()

# ------------------------------
# Elbow Method
# ------------------------------

features_el = df.iloc[:, [2, 3, 4]].values

wcss = []

for i in range(1, 10):
    model = KMeans(n_clusters=i, n_init=10, random_state=42)
    model.fit(features_el)
    wcss.append(model.inertia_)

# Plot WCSS
plt.plot(range(1, 10), wcss, marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64

dtypes: int64(4), object(1)
memory usage: 7.9+ KB

First five rows:

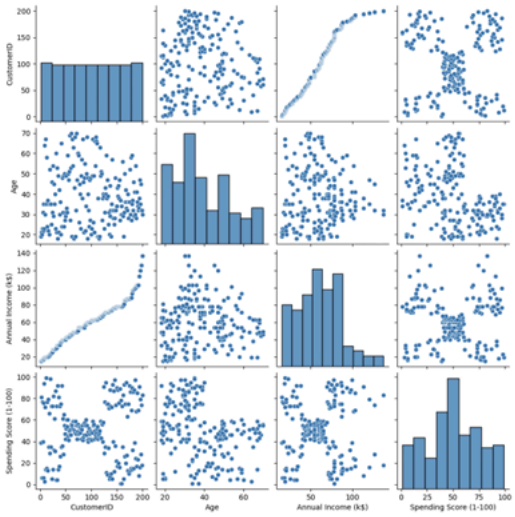
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

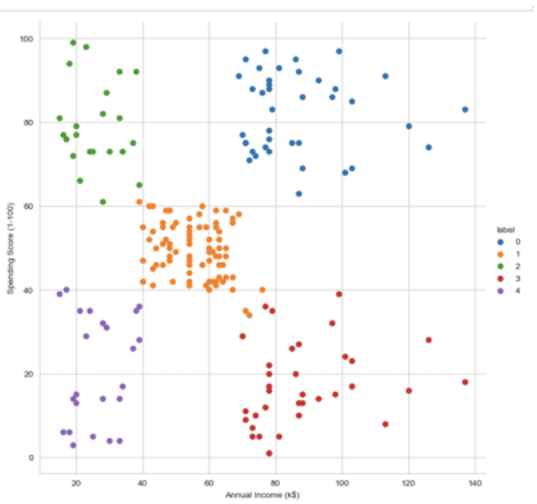
KMeans(n_clusters=5)

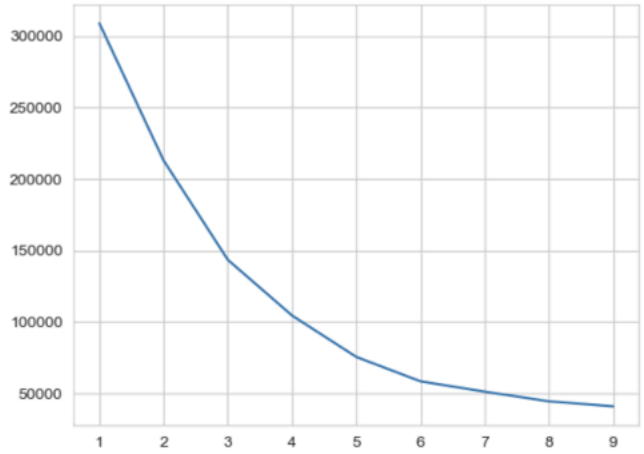
First five clustered records:

   Annual Income (k$)  Spending Score (1-100)  label
0                  15                      39      4
1                  15                      81      2
2                  16                       6      4
3                  16                      77      2
4                  17                      40      4

Plots produced:
1. Pair plot of the dataset
2. K-Means customer-cluster scatter plot
3. Elbow-method WCSS plot for clusters 1 to 9
```







RESULT:

The code is executed successfully.